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INTRUSION DETECTION AND PREVENTION SYSTEMS (IDPS)

NETWORK SECURITY

OBJECTIVES

- ▶ IDPS Introduction
- ▶ Types of IDPS
- ▶ Features of IDPS
- ▶ Snort



WHAT IS IDPS

- ▶ Intrusion Detection and Prevention Systems (IDPS) is a network security solution that monitors, detects, and prevents potential malicious activity or policy violations.
- ▶ Functions of IDPS:
 - Monitors network traffic in real-time
 - Detects intrusions based on predefined rules or behavior
 - Prevents or mitigates attacks by blocking or alerting

TYPES OF IDPS

- ▶ **Network-Based IDPS (NIDPS):** Monitors traffic at the network level, identifies suspicious patterns, and reacts to prevent attacks.
- ▶ **Host-Based IDPS (HIDPS):** Operates on individual devices or hosts, monitoring activity like file changes or logins.
- ▶ **Signature-Based Detection:** Matches known patterns of malicious activity (signatures).
- ▶ **Anomaly-Based Detection:** Monitors deviations from normal activity to identify potential threats.

KEY FEATURES OF IDPS – REAL-TIME TRAFFIC ANALYSIS

- ▶ Real-time traffic analysis is a core feature of IDPS that allows it to monitor network traffic as it flows through the network, detecting suspicious activities as they happen. This feature involves:
 - ▶ **Continuous Monitoring:** Inspects packets that pass through the network in real-time. Checks headers, payloads, and other packet attributes to detect anomalies or known attack patterns.
 - ▶ **Pattern Matching:** Compares network traffic against a set of predefined rules or signatures, such as known attack signatures (e.g., viruses, exploits) or suspicious behavior (e.g., unusual bandwidth usage, abnormal connection requests).
 - ▶ **Protocol Analysis:** Inspects traffic for correct protocol usage. For example, an IDPS can detect when HTTP traffic (usually TCP port 80) is using an unexpected port.

KEY FEATURES OF IDPS – ALERTING AND LOGGING

- ▶ **Alert Generation:** When the IDPS detects suspicious activity that matches its rules or signatures, it triggers an alert. These alerts typically contain important information such as:
 - ▶ Type of attack or activity (e.g., port scanning, SQL injection attempt)
 - ▶ Source and destination IP addresses
 - ▶ The exact rule or signature that was triggered
 - ▶ Timestamp of the event
- ▶ **Real-Time Alerts:** Many IDPS solutions allow real-time alerts via email, SMS, or integration with security information and event management (SIEM) systems.

KEY FEATURES OF IDPS – ALERTING AND LOGGING CONT.

- ▶ **Logging:** All detected activities and their details are logged for later review. These logs are crucial for post-incident analysis, auditing, and compliance purposes. They contain data such as:
 - Full details of the traffic involved in the alert (e.g., packet headers, payloads)
 - Historical information on incidents for pattern recognition or forensic investigation
 - Time-stamped entries that help establish timelines for security incidents

KEY FEATURES OF IDPS – THREAT PREVENTION

- ▶ In Intrusion Prevention System (IPS) mode, your IDPS can:
- ▶ **Block Malicious Traffic:** When malicious traffic is detected, the IDPS can automatically drop packets, block IP addresses, or close connections that are deemed dangerous.
- ▶ **Throttle Traffic:** For threats like DDoS attacks, an IDPS can limit the rate of traffic flow from suspected attackers, slowing down or preventing the overload of network resources.
- ▶ **Inline Mode:** Operates between the network's entry point and critical systems, analyzing packets before they reach their destination. This placement allows the system to intercept, analyze, and block malicious traffic before it harms internal resources. Acts as a gatekeeper for your network.

KEY FEATURES OF IDPS – PACKET CAPTURE

- ▶ **Deep Packet Inspection (DPI):** Can inspect the actual data payload to identify malicious activities like SQL injections, command injections, or buffer overflow attempts.
- ▶ **Full Packet Capture:** Can store a full copy of network packets. Valuable for forensic analysis, allowing administrators to review exactly what happened during an attack and potentially reconstruct the entire event.
- ▶ **Packet Analysis Tools:** Many IDPS solutions integrate with packet analysis tools such as Wireshark or tcpdump. These tools allow for detailed analysis of captured traffic, helping security teams understand how attacks occurred and what data may have been compromised.

KEY FEATURES OF IDPS – INCIDENT RESPONSE INTEGRATION

- ▶ **Automated Response Actions:** Can be configured to take automatic actions when a threat is detected, such as:
 - Quarantining affected systems
 - Isolating malicious traffic from the network
 - Blocking the IP address or domain associated with the attack
- ▶ **SIEM Integration:** IDPS solutions often integrate with Security Information and Event Management (SIEM) systems.
- ▶ **Incident Reporting:** The detailed logging and alerting capabilities of an IDPS provide incident responders with vital information about detected threats, including attack patterns, affected systems, and the timeline of events. Helps understand the scope of the attack and plan remediation steps.

KEY FEATURES OF IDPS – INCIDENT RESPONSE INTEGRATION CONT.

- ▶ **Post-Incident Forensics:** IDPS logs and packet captures serve as essential evidence for post-incident investigations. The captured data allows incident responders to:
 - Trace the origin of the attack
 - Identify the methods used by the attackers
 - Determine whether data was exfiltrated or modified

WHAT IS SNORT

- ▶ **Snort** is an open-source IDPS developed by Cisco. It provides real-time traffic analysis and packet logging, capable of detecting and preventing a wide range of attacks.
- ▶ Snort Features:
 - Network traffic monitoring
 - Deep packet inspection
 - Rule-based analysis
 - Detection of attacks like buffer overflows, port scans, and more

WHAT CAN SNORT DO

- ▶ **Packet Capture:** Snort captures network packets for analysis.
- ▶ **Preprocessing:** Prepares the packets for inspection by reassembling fragmented data and normalizing content.
- ▶ **Detection Engine:** Compares packets against rules to detect suspicious activities.
- ▶ **Logging and Alerting:** Logs malicious activity and sends alerts to system administrators.
- ▶ **Prevention (if configured as IPS):** Blocks malicious traffic in real-time.

SNORT RULE SYNTAX

```
alert tcp any any -> 192.168.1.0/24 80 (msg:"Possible HTTP Attack"; sid:1000001;)
```

- ▶ Action: alert
- ▶ Protocol: tcp
- ▶ Source/Destination IP and Port: 192.168.1.0/24 80
- ▶ Message: Provides details of the alert
- ▶ SID (Unique Snort rule identifier): 1000001
- ▶ <https://docs.snort.org/rules/headers/actions>

ADVANTAGES OF SNORT

- ▶ Open-Source and Free
- ▶ Flexible and Configurable
- ▶ Large Community Support
- ▶ Scalable for Networks of Any Size
- ▶ Regular Updates for Threat Signatures

CHALLENGES AND LIMITATIONS

- ▶ **False Positives:** May generate alerts for legitimate traffic.
- ▶ **Complex Rule Management:** Requires regular updates and tuning of rules.
- ▶ **Resource Intensive:** High traffic volumes may require significant system resources.

IDPS – ASSIGNMENT 6

- ▶ Read from **Introduction** to **Rule Headers**: <https://docs.snort.org/welcome>
- ▶ Complete the TryHackMe lab on Snort: <https://tryhackme.com/r/room/snort>
- ▶ Set up Snort on your VM (directions in PDF under latest module).